Predictive Modeling of Brain Stroke Risk Using Machine Learning Techniques

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Abstract

Stroke is a major global health issue, making early prediction important in healthcare research. This study aimed to predict stroke occurrence using demographic, medical, and lifestyle-related variables associated with stroke risk. Several preprocessing techniques, including duplicate removal, outlier detection, one-hot encoding, standardization, and Synthetic Minority Oversampling Technique (SMOTE) balancing, were applied before analysis. Exploratory Data Analysis (EDA) was conducted to identify important relationships and stroke-related risk factors. Three machine learning models Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) were implemented and evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix analysis. The findings identified age, average glucose level, hypertension, and BMI as important predictors of stroke occurrence. Among the models, Random Forest achieved the highest accuracy of 89.67%, while Logistic Regression demonstrated the best ROC-AUC performance with a value of 0.84. The KNN model achieved moderate predictive performance compared to the other models. Overall, the study demonstrates the usefulness of machine learning techniques in supporting early stroke prediction and identifying high-risk patients.

Content

List of Figures and Tables

List of Tables

List of Figures

01. Introduction

Stroke is one of the leading causes of death and long-term disability worldwide and occurs when the blood supply to the brain is interrupted, preventing brain cells from receiving enough oxygen and nutrients. Factors such as hypertension, heart disease, high glucose levels, obesity, smoking, and aging significantly increase stroke risk. Since many of these factors can be prevented or controlled, early identification of high-risk individuals is important for reducing mortality and improving healthcare outcomes.

With the growth of healthcare data and advancements in machine learning, predictive analytics has become an effective tool in medical decision-making. Machine learning techniques can analyze large datasets, identify hidden patterns, and predict diseases with improved accuracy. In stroke prediction, these methods help healthcare professionals identify individuals who may be at risk before severe symptoms occur, supporting early diagnosis and preventive healthcare strategies.

This study focuses on the predictive analysis of brain stroke risk using demographic, medical, and lifestyle-related variables. The dataset includes variables such as gender, age, marital status, residence type, hypertension, heart disease, average glucose level, BMI, work type, and smoking status, while stroke occurrence is considered as the target variable.

Objectives of the Study:

• To develop machine learning classification models for predicting brain stroke risk based on patient characteristics.

• To identify the most significant factors contributing to brain stroke occurrence.

The study includes several stages of analysis such as data preprocessing, which involves cleaning and preparing the dataset for analysis; exploratory data analysis, which examines patterns and relationships among variables; feature selection, which identifies the most relevant factors influencing stroke occurrence; handling class imbalance, which ensures fair representation of stroke and non-stroke cases; model building, which develops predictive classification models; and model evaluation, which assesses how effectively the models identify stroke risk using performance measures.

The findings of this study are expected to support healthcare professionals and researchers in identifying individuals with a higher risk of stroke and improving early preventive measures. In addition, understanding the influence of factors such as age, hypertension, glucose level, BMI, and smoking habits can help increase public awareness and support better healthcare planning and decision-making.

02. Literature Review

Stroke remains a leading cause of death and long-term disability globally, with early detection and risk prediction being critical to improving patient outcomes. Recent advances in machine learning (ML) have enabled the development of predictive models that can assist healthcare providers in identifying high-risk individuals based on demographic, medical, and lifestyle characteristics.

Dritsas and Trigka (2022) developed and compared machine learning models for stroke risk prediction using Logistic Regression, Random Forest, and Naive Bayes with variables such as age, hypertension, heart disease, glucose level, and BMI. Their findings showed that Random Forest achieved the highest predictive accuracy by effectively handling nonlinear relationships and class imbalance. Logistic Regression provided moderate accuracy with better interpretability, while Naïve Bayes showed limitations when dealing with correlated features.

Hassan et al. (2024) examined stroke risk prediction using Logistic Regression, Decision Tree, and Random Forest classifiers, confirming that age, average glucose level, and hypertension were the strongest predictors of stroke occurrence across all models. Random Forest again outperformed the other models in overall accuracy and sensitivity. The study also demonstrated that applying SMOTE oversampling to address severe class imbalance significantly improved model recall and F1-score, particularly for correctly identifying minority stroke cases within the dataset.

Vu et al. (2024) applied Logistic Regression, Random Forest, and Support Vector Machine (SVM) to stroke risk prediction using longitudinal clinical patient data. Their comparative analysis confirmed that Random Forest consistently outperformed simpler models in sensitivity and overall predictive accuracy. The study further emphasized the importance of incorporating diverse demographic, medical, and lifestyle variables, which closely mirrors the variable structure of the dataset used in the present study.

Alizadeh-dizaj and Khoshsirat (2025) investigated stroke risk classification using Logistic Regression, K-Nearest Neighbors, and Random Forest. Their study found that Random Forest achieved the best overall performance after applying preprocessing and SMOTE balancing techniques. K-Nearest Neighbors performed reasonably on balanced data but was sensitive to irrelevant features and less effective on larger datasets.

Collectively, these studies highlight the trade-offs between interpretability and predictive performance. Logistic Regression offers transparency but struggles with complex, imbalanced data patterns. K-Nearest Neighbors is straightforward but sensitive to noise and irrelevant features. Random Forest, though less interpretable, consistently delivers superior accuracy across all reviewed studies when handling imbalanced medical datasets. This insight guided the selection and comparative evaluation of Logistic Regression, Random Forest, and K-Nearest Neighbors in the present study.

03. Theory and Methodology

This section presents the theoretical background and methodology used to predict brain stroke occurrence using machine learning techniques. The analysis was conducted using Python programming language together with several data preprocessing, exploratory analysis, and predictive modelling techniques. The methodology was designed to ensure accurate data preparation, identify important stroke-related risk factors, and evaluate the predictive performance of different machine learning models.

- **Data Loading and Initial Inspection**

The dataset used in this study was obtained from Kaggle and imported into Python using the Pandas library. Initial inspection was conducted to understand the dataset structure, variable types, missing values, and duplicate records. The dataset contained demographic, medical, and lifestyle-related variables relevant to stroke prediction. Duplicate observations were identified and removed to improve data quality and ensure reliable analysis.

- **Exploratory Data Analysis (EDA)**

Exploratory Data Analysis (EDA) was conducted to understand variable distributions and relationships with the response variable, stroke. Univariate analysis used histograms, boxplots, and bar charts to visualize quantitative and qualitative variables individually. Bivariate analysis was performed to examine relationships between predictor variables and stroke occurrence using boxplots and grouped bar charts. A correlation heatmap was also used to identify relationships among numerical variables and determine important predictors associated with stroke occurrence.

- **Outlier Detection and Removal**

During EDA, several outliers were identified, particularly in the variables average glucose level and BMI. Since extreme values can negatively affect machine learning performance, the Interquartile Range (IQR) method was applied to detect and remove outliers.

The IQR was calculated as:

$$IQR = Q_3 - Q_1$$

The lower and upper limits were calculated using:

$$Lower\ Bound = Q_1 - 1.5(IQR)$$
$$Upper\ Bound = Q_3 + 1.5(IQR)$$

Observations outside these boundaries were treated as outliers and removed from the dataset to improve model reliability and reduce noise within the data.

- **Handling Missing Values**

The dataset was examined for missing values using Python functions such as isnull() and sum(). No significant missing values were detected in most variables. However, the smoking status variable included categories such as “Unknown,” which were handled appropriately during preprocessing to avoid bias in the analysis.

Proper handling of missing information is important because incomplete data may reduce predictive accuracy and affect the validity of machine learning models.

- **Encoding Categorical Variables**

Categorical variables such as gender, work type, residence type, smoking status, and marital status were converted into numerical format using one-hot encoding. This technique creates separate binary variables for each category and prevents machine learning models from incorrectly interpreting categorical values as ordinal data.

- **Standardizing the Data**

The quantitative variables age, average glucose level, and BMI were standardized using the StandardScaler method from the Scikit-learn library. Standardization transformed the variables to have a mean of 0 and a standard deviation of 1.The standardized value was calculated using:

$$Z = \frac{X - \mu}{\sigma}$$

- Z= standardized value
- X= original value
- μ= mean
- σ= standard deviation

This step was particularly important for the K-Nearest Neighbors (KNN) model because distance-based algorithms are sensitive to differences in variable scales.

- **Splitting the Dataset**

The dataset was divided into training and testing datasets using an 80:20 ratio. The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate predictive performance on unseen observations.Stratified sampling was applied during the splitting process to preserve the proportion of stroke and non-stroke cases in both datasets. This approach ensured more reliable and unbiased model evaluation.

- **Balancing the Response Variable**

The response variable stroke was highly imbalanced because non-stroke cases significantly outnumbered stroke cases. To reduce classification bias toward the majority class, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training dataset. SMOTE generated synthetic samples for the minority class, improving the ability of the models to identify stroke-positive cases more effectively.

- **Logistic Regression Model**

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. The model estimates the probability of stroke occurrence using the logistic sigmoid function.

The Logistic Regression equation is represented as:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0+\beta_1 X_1+\beta_2 X_2+\cdots+\beta_n X_n)}}$$

- $P(Y = 1)$= probability of stroke occurrence
- β_0= intercept
- $\beta_1, \beta_2, \ldots, \beta_n$= regression coefficients
- $X_1, X_2, \ldots, X_n$= predictor variables

Logistic Regression was selected because it is interpretable, computationally efficient, and suitable for medical classification problems involving binary outcomes.

- **Random Forest Model**

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting. The final prediction is determined through majority voting among all trees in the model.

The Random Forest prediction function is represented as:

$$\hat{y} = mode\{h_1(x), h_2(x), \ldots, h_T(x)\}$$

- $\hat{y}$= predicted class
- $h_t(x)$= prediction from tree t
- T= total number of trees

Random Forest was selected because it performs well with complex and nonlinear relationships and can effectively identify important predictor variables.

- **K-Nearest Neighbors (KNN) Model**

K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies observations based on the similarity of neighboring data points. A new observation is assigned to the majority class among its nearest neighbors.

The Euclidean distance formula used in KNN is:

$$d(p, q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$$

- $d(p, q)$= distance between points $p$and q
- p_i= value of observation $p$for feature i
- q_i= value of observation $q$for feature i
- n= number of variables

KNN was selected because it is simple, flexible, and effective for identifying local patterns within the dataset.

- **Model Evaluation Techniques**

The machine learning models were evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, Classification Report, and Confusion Matrix analysis.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2(Precision \times Recall)}{Precision + Recall}$$

where TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

ROC-AUC analysis was used to evaluate the classification ability of each model, while confusion matrices summarized correct and incorrect classifications. Finally, overall model comparison was performed to identify the best-performing model for stroke prediction.

04. Data Description

The dataset used in this study was obtained from the https://www.kaggle.com/datasets/jillanisofttech/brain-stroke-dataset and contains demographic, medical, and lifestyle-related information associated with stroke occurrence. The dataset consisted of 4,981 patient records with 11 variables representing several health conditions and personal characteristics relevant to stroke prediction. The target variable of the study was *stroke*, a binary variable indicating whether a patient experienced a stroke (1 = Yes, 0 = No).

The dataset included both quantitative and qualitative variables. The quantitative variables were age, average glucose level, and BMI, while the qualitative variables included gender, hypertension status, heart disease status, marital status, work type, residence type, smoking status, and stroke occurrence. These variables were selected because they represent important demographic and medical risk factors commonly associated with stroke prediction.

Several preprocessing techniques were applied before conducting the analysis to improve data quality and model performance. Initial inspection was performed to identify missing values, duplicate observations, and variable types. Duplicate records were removed to ensure dataset consistency and reliability. Outliers identified mainly in the average glucose level and BMI variables were detected using the Interquartile Range (IQR) method and handled appropriately to reduce the impact of extreme values on predictive modelling.

Since machine learning algorithms require numerical input, categorical variables were transformed into numerical format using one-hot encoding. In addition, numerical variables were standardized using the StandardScaler technique to maintain consistency among variable scales and improve model efficiency, particularly for distance-based algorithms such as K-Nearest Neighbors (KNN). Furthermore, because the response variable stroke was highly imbalanced, the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the minority class and improve the models' ability to identify stroke-positive cases more effectively.

Table 4.1 presents the variables used in this study together with their variable types and descriptions. The dataset contains both quantitative and qualitative variables related to demographic, medical, and lifestyle characteristics that may influence stroke occurrence.

Variable Name	Description	Variable Type
age	Age of the patient	Quantitative
avg_glucose_level	Average glucose level in the patient’s blood	Quantitative
bmi	Body Mass Index of the patient	Quantitative
gender	Gender of the patient	Qualitative
hypertension	Hypertension status of the patient (1 = Yes, 0 = No)	Qualitative
heart_disease	Presence of heart disease (1 = Yes, 0 = No)	Qualitative
ever_married	Marital status of the patient	Qualitative
work_type	Type of occupation of the patient	Qualitative
Residence_type	Residence type of the patient (Urban/Rural)	Qualitative
smoking_status	Smoking behavior of the patient	Qualitative
stroke	Stroke occurrence status (1 = Yes, 0 = No)	Qualitative

Table 4.1:Description of Variables Used in the Study

05. Exploratory Data Analysis (EDA)

This section presents an exploration of the brain stroke dataset to understand the distributions, relationships, and characteristics of the variables before model development. The dataset contains demographic, medical, and lifestyle-related information relevant to stroke prediction, including variables such as age, gender, hypertension, heart disease, marital status, work type, residence type, average glucose level, BMI, and smoking status. Both univariate and bivariate analyses were conducted using descriptive statistics and visualization techniques such as histograms, boxplots, bar charts, and correlation heatmaps to identify patterns, trends, and associations with stroke occurrence.

Univariate Analysis

Quantitative Variables

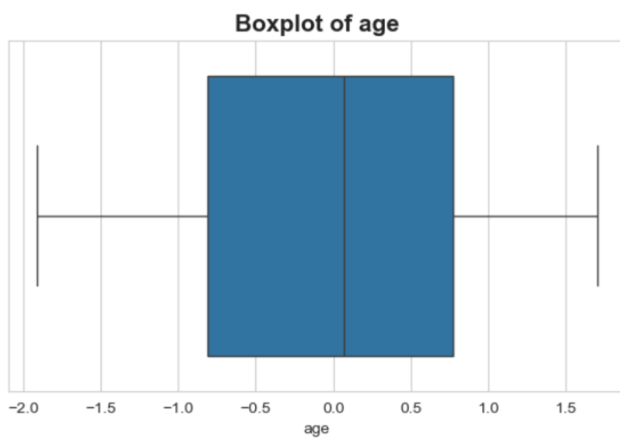


Figure 5.1: Boxplot of the Standardized Age Variable

Figure 5.1 shows that the standardized age variable is approximately symmetrically distributed, with a moderate spread of values and no significant outliers. This indicates that the age data are relatively well balanced without extreme variability.

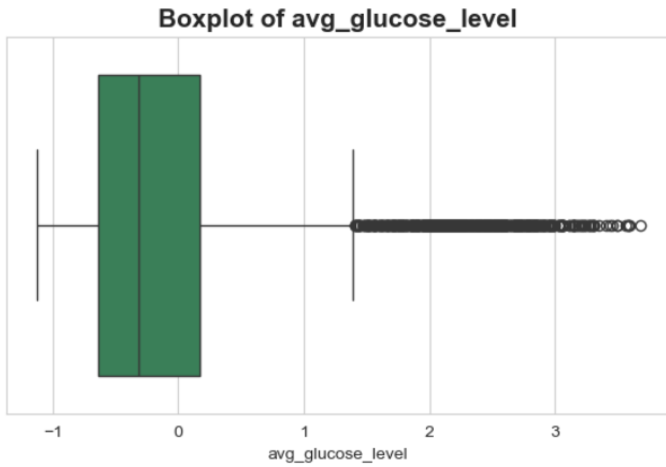


Figure 5.2 : Boxplot of the Standardized Average Glucose Level Variable

Figure 5.2 shows that the standardized average glucose level variable is positively skewed, with several high-value outliers observed on the right side of the distribution. The median is slightly closer to the lower quartile, indicating that most observations are concentrated at lower glucose levels. This suggests the presence of extreme glucose values among some individuals in the dataset.

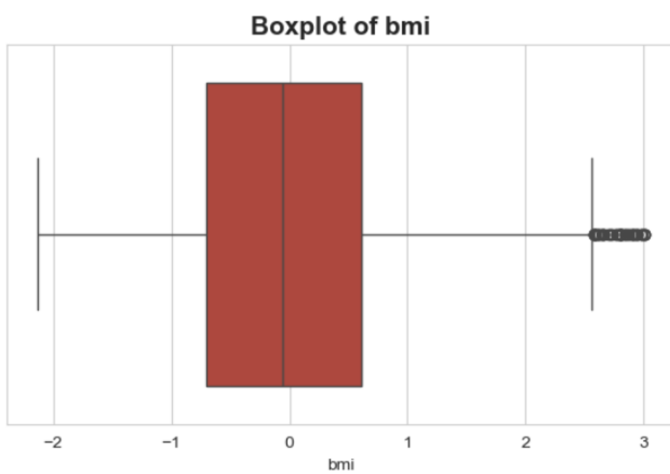


Figure 5.3 :Boxplot of the Standardized Body Mass Index (BMI) Variable

Figure 5.3 indicates that the standardized BMI variable is slightly positively skewed, with several outliers appearing at higher BMI values. The median is located near the center of the box, suggesting a relatively balanced distribution for most observations. However, the presence of outliers indicates that some individuals have unusually high BMI levels compared to the rest of the dataset.

Qualitative Variables

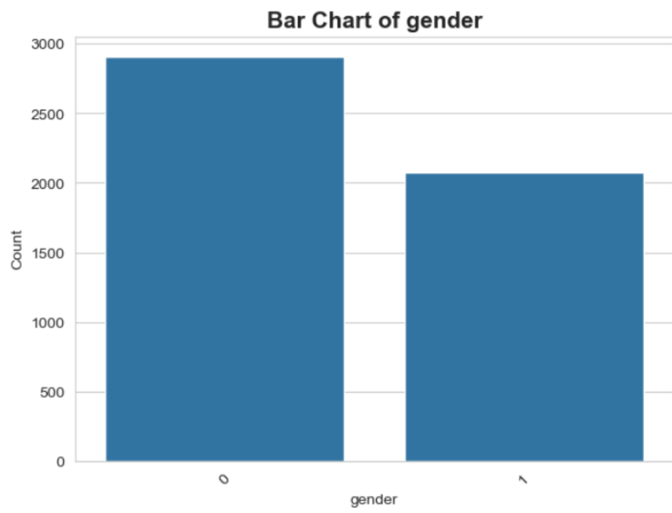


Figure 5.4: Distribution of Gender

Note. Gender was encoded as Female = 0 and Male = 1.

Figure 5.4 illustrates the distribution of gender within the dataset. The category labeled “0” (Female) has a higher number of observations compared to the category labeled “1” (Male), indicating that female participants are more represented in the dataset than male participants.

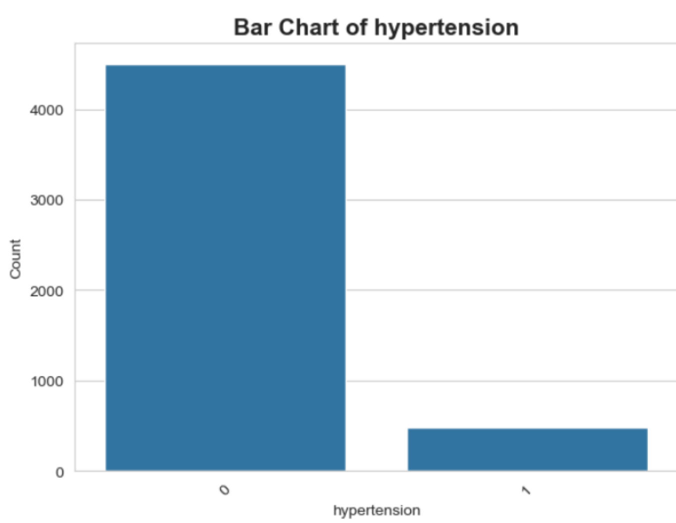


Figure 5.5: Distribution of Hypertension Status

Note. Hypertension was encoded as No Hypertension = 0 and Hypertension = 1.

Figure 5.5 illustrates the distribution of hypertension status within the dataset. The category labeled “0” (No Hypertension) contains a substantially higher number of observations compared to the category labeled “1” (Hypertension), indicating that most individuals in the dataset do not have hypertension.

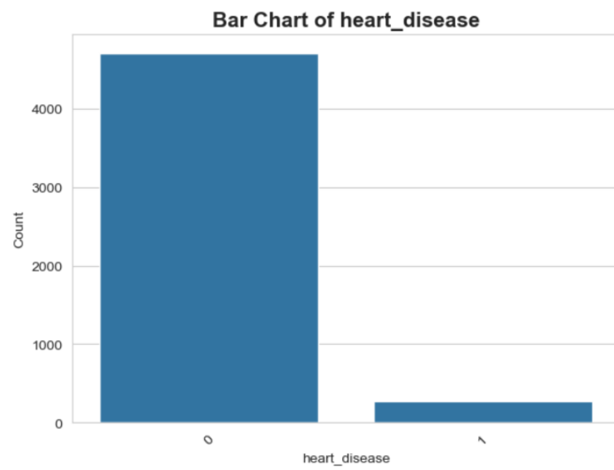


Figure 5.6:*Distribution of Heart Disease Status*

Note. Heart disease status was encoded as No Heart Disease = 0 and Heart Disease = 1.

Figure 5.6 illustrates the distribution of heart disease status within the dataset. The category labeled “0” (No Heart Disease) contains a considerably higher number of observations compared to the category labeled “1” (Heart Disease), indicating that the majority of individuals in the dataset do not have heart disease.

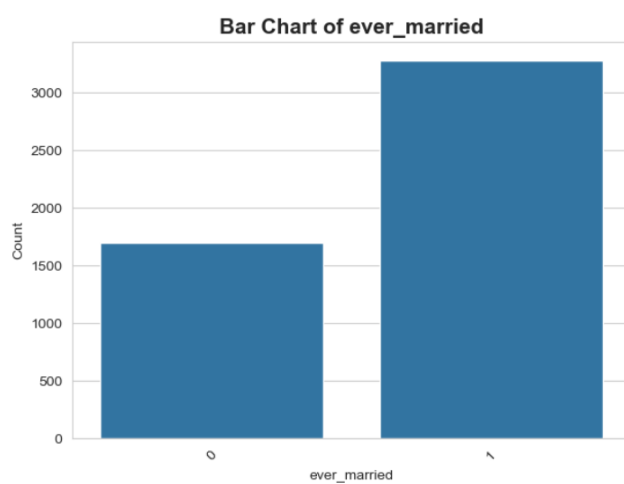


Figure 5.7:Distribution of Marital Status

Note. Marital status was encoded as Not Married = 0 and Ever Married = 1.

Figure 5.7 illustrates the distribution of marital status within the dataset. The category labeled "1" (Ever Married) contains a higher number of observations compared to the category labeled "0" (Not Married), indicating that most individuals have been married.

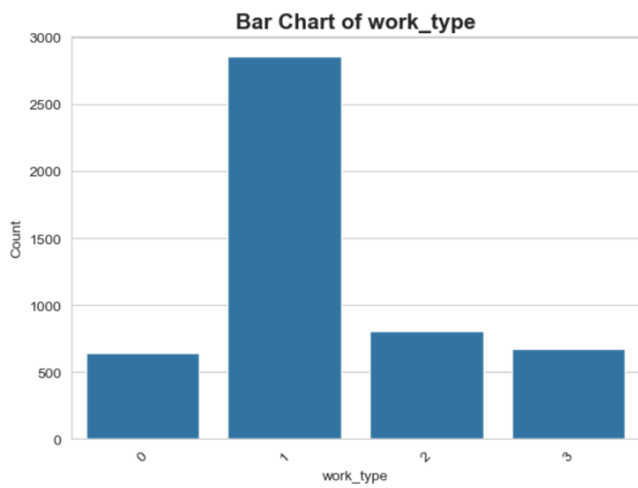


Figure 5.8:Distribution of Work Type

Note. Work type was encoded as Children = 0, Private = 1, Self-employed = 2, and Government Job = 3.

Figure 5.8 illustrates the distribution of work type within the dataset. The category labeled "1" (Private) contains the highest number of observations compared to the other work categories, indicating that most individuals in the dataset are employed in the private sector. The remaining categories show relatively lower frequencies.

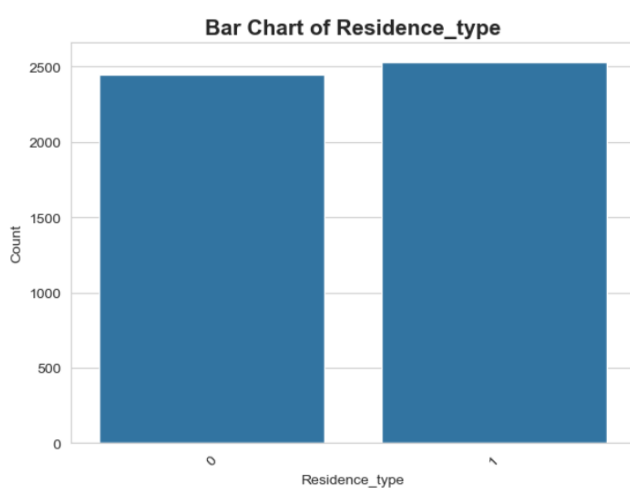


Figure 5.9: Distribution of Residence Type

Note. Residence type was encoded as Rural = 0 and Urban = 1.

Figure 5.9 illustrates the distribution of residence type within the dataset. The frequencies of the two categories are relatively similar, although the category labeled “1” (Urban) contains a slightly higher number of observations compared to the category labeled “0” (Rural). This indicates that urban residents are marginally more represented in the dataset.

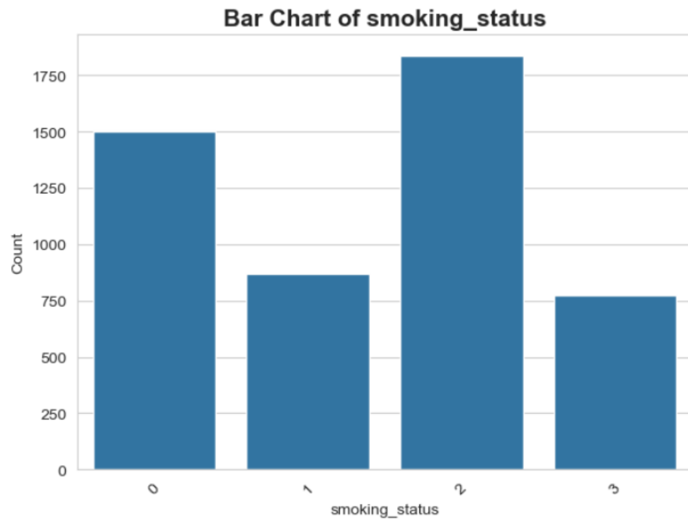


Figure 5.10: *Distribution of Smoking Status*

Note. Smoking status was encoded as Formerly Smoked = 0, Never Smoked = 1, Smokes = 2, and Unknown = 3.

Figure 5.10 illustrates the distribution of smoking status within the dataset. The category labeled “2” (Smokes) contains the highest number of observations compared to the other categories, indicating that smokers are more represented in the dataset. The categories “0” (Formerly Smoked), “1” (Never Smoked), and “3” (Unknown) show comparatively lower frequencies.

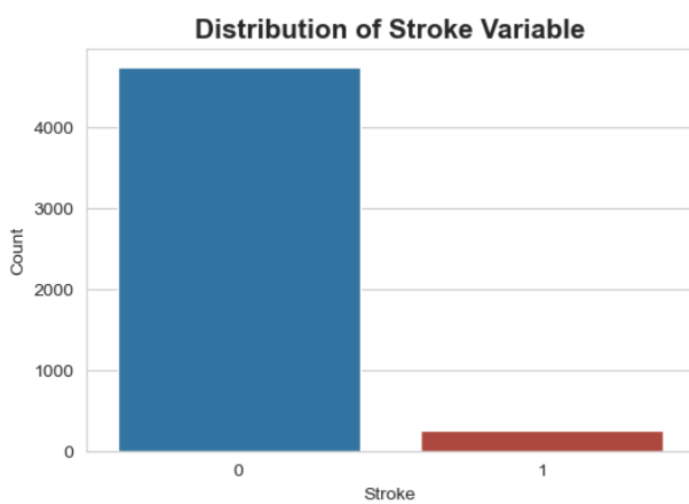


Figure 5.11: Distribution of Stroke Occurrence

Note. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.11 illustrates the distribution of stroke occurrence within the dataset. The category labeled “0” (No Stroke) contains a substantially higher number of observations compared to the category labeled “1” (Stroke), indicating that the dataset is highly imbalanced with relatively fewer stroke cases.

Bivariate Analysis

Quantitative Variables vs Stroke

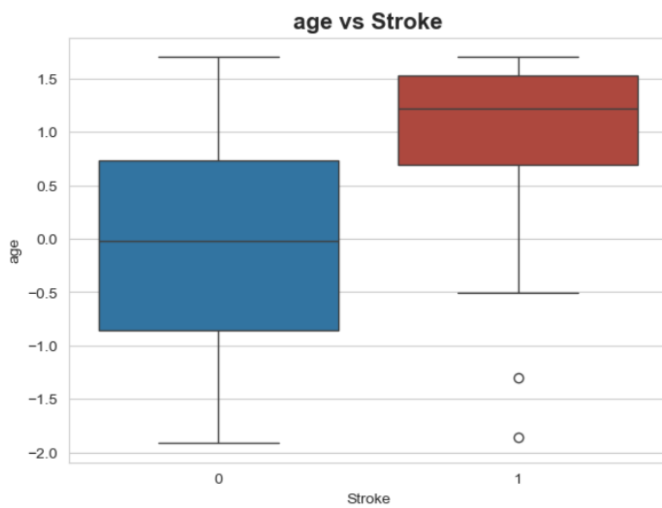


Figure 5.12: Age Distribution by Stroke Occurrence

Note. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.12 illustrates the distribution of age according to stroke occurrence. Individuals in the stroke group (1) generally show higher age values compared to individuals in the non-stroke group (0), indicating that stroke occurrence is more common among older individuals. A few outliers are also observed within the stroke group.

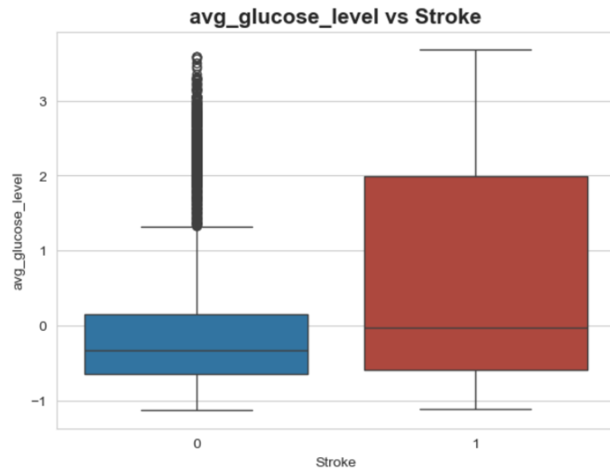


Figure 5.13: Average Glucose Level Distribution by Stroke Occurrence

Note. Stroke status was encoded as No Stroke = 0 and Stroke = 1. The average glucose level variable was standardized before visualization.

Figure 5.13 illustrates the distribution of average glucose levels according to stroke occurrence. Individuals in the stroke category (“1”) show higher glucose level values and greater variability compared to individuals in the non-stroke category (“0”). Several outliers are also observed in the non-stroke group, indicating the presence of unusually high glucose levels among some individuals.

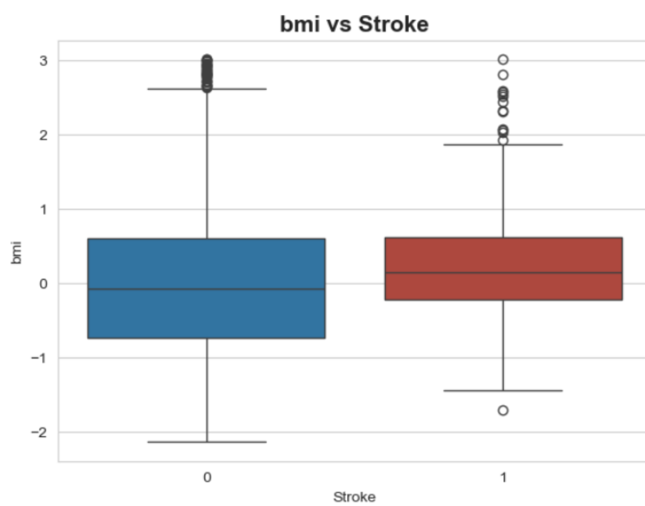


Figure 5.14: BMI Distribution by Stroke Occurrence

Note. Stroke status was encoded as No Stroke = 0 and Stroke = 1. The BMI variable was standardized before visualization.

Figure 5.14 illustrates the distribution of BMI according to stroke occurrence. Individuals in the stroke category (“1”) show a slightly higher median BMI compared to individuals in the non-stroke category (“0”). Both groups contain several outliers, indicating the presence of unusually high BMI values within the dataset.

Qualitative Variables vs Stroke

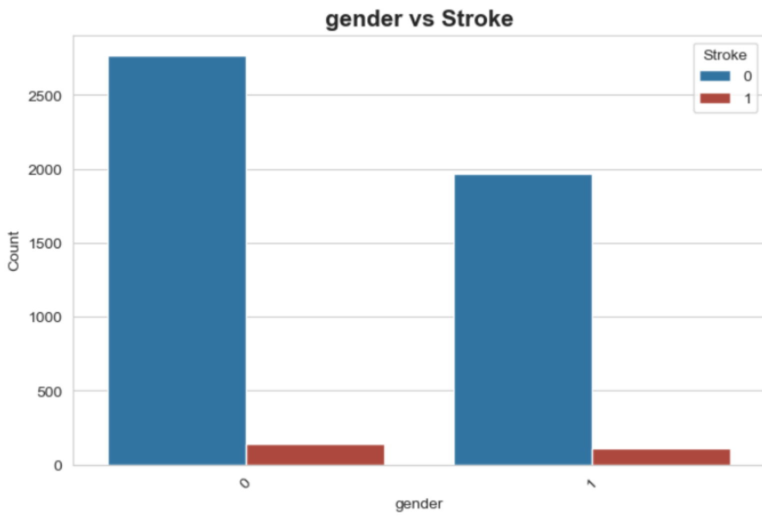


Figure 5.15: Gender Distribution by Stroke Occurrence

Note. Gender was encoded as Female = 0 and Male = 1. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.15 illustrates the distribution of stroke occurrence according to gender. In both gender categories, the number of individuals without stroke ("0") is substantially higher than the number of individuals with stroke ("1"). The female category ("0") contains a slightly higher number of observations compared to the male category ("1"), indicating a greater representation of females in the dataset.

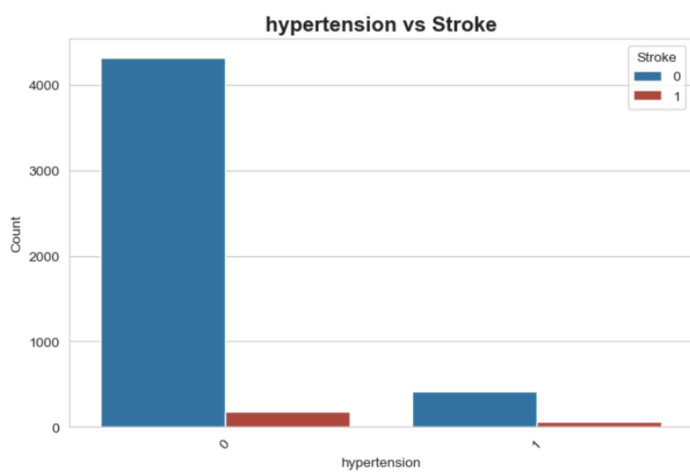


Figure 5.16: Hypertension Status by Stroke Occurrence

Note. Hypertension status was encoded as No Hypertension = 0 and Hypertension = 1. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.16 presents the relationship between hypertension status and stroke occurrence. Individuals without hypertension ("0") account for the majority of observations in the dataset. However, the proportion of stroke cases appears relatively higher among individuals with hypertension ("1"), suggesting that hypertension may be associated with an increased likelihood of stroke occurrence.

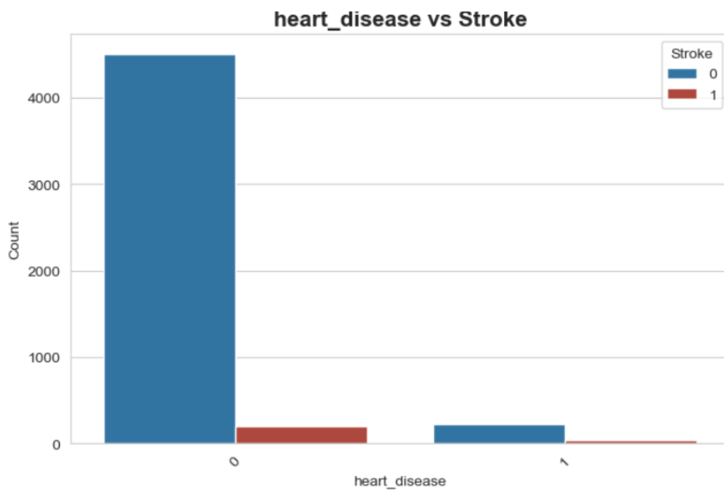


Figure 5.17: Heart Disease Status by Stroke Occurrence

Note. Heart disease status was encoded as No Heart Disease = 0 and Heart Disease = 1. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.17 displays the relationship between heart disease status and stroke occurrence. Most individuals in the dataset belong to the category without heart disease ("0"). Although the number of individuals with heart disease ("1") is comparatively smaller, the presence of stroke cases within this group suggests a possible association between heart disease and stroke risk.

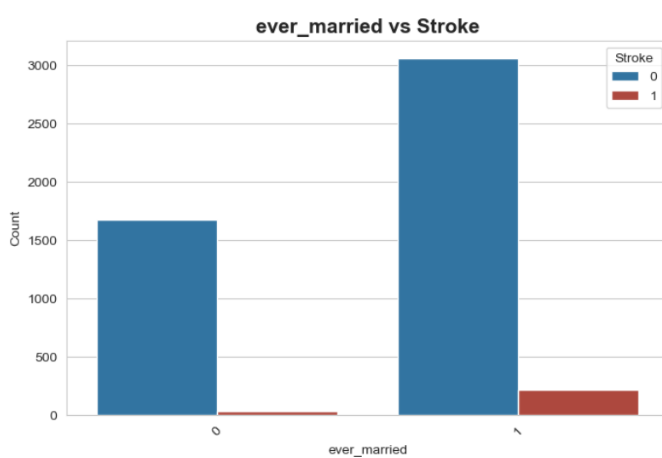


Figure 5.18: Marital Status by Stroke Occurrence

Note. Marital status was encoded as Not Married = 0 and Ever Married = 1. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.18 illustrates the association between marital status and stroke occurrence. The majority of individuals in the dataset belong to the "Ever Married" category ("1"). In addition, stroke cases are more frequently observed among individuals who have been married compared to those who have not been married, indicating a potential relationship between marital status and stroke occurrence.

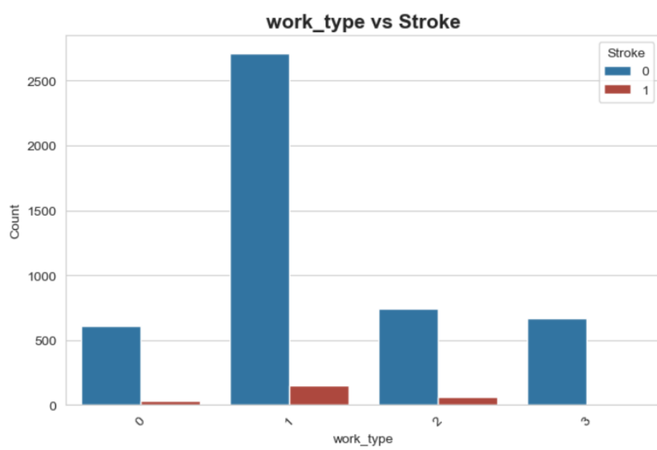


Figure 5.19: Work Type by Stroke Occurrence

Note. Work type was encoded as Children = 0, Private = 1, Self-employed = 2, and Government Job = 3. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.19 presents the distribution of stroke occurrence across different work types. The "Private" work category ("1") contains the largest number of individuals as well as the highest number of stroke cases compared to the other categories. In contrast, the "Government Job" category ("3") shows relatively fewer observations and very few stroke cases. These patterns suggest that stroke occurrence varies across employment categories within the dataset.

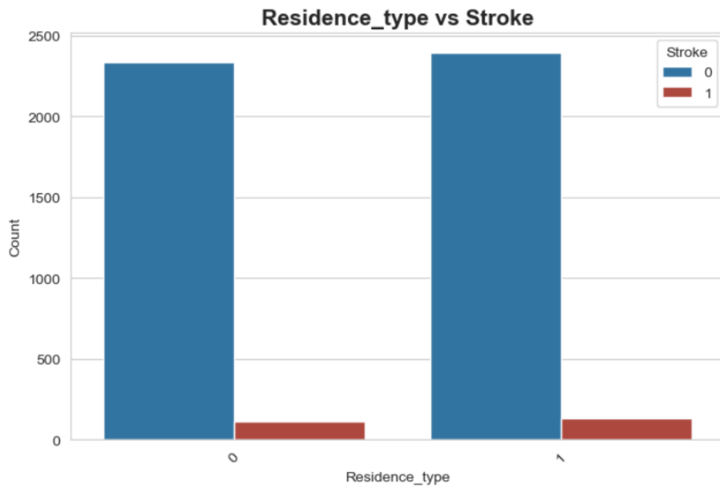


Figure 5.20: Residence Type by Stroke Occurrence

Note. Residence type was encoded as Rural = 0 and Urban = 1. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.20 shows the relationship between residence type and stroke occurrence. The numbers of individuals from rural (“0”) and urban (“1”) areas are relatively similar within the dataset. Stroke cases are observed in both residence categories, with a slightly higher frequency among urban residents, indicating a minor difference in stroke occurrence based on residence type.

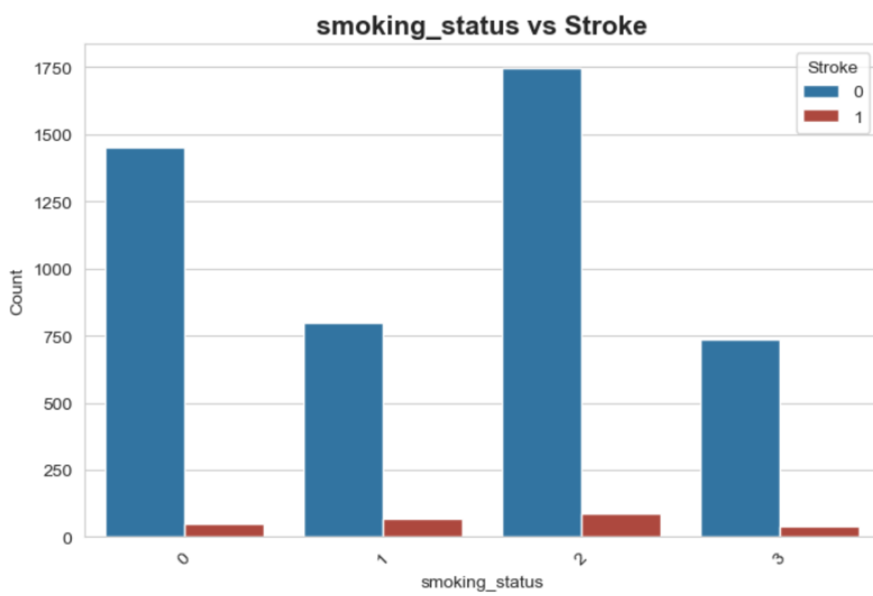


Figure 5.21: Smoking Status by Stroke Occurrence

Note. Smoking status was encoded as Formerly Smoked = 0, Never Smoked = 1, Smokes = 2, and Unknown = 3. Stroke status was encoded as No Stroke = 0 and Stroke = 1.

Figure 5.21 illustrates the association between smoking status and stroke occurrence. The “Smokes” category (“2”) contains the highest number of individuals as well as the greatest number of stroke cases compared to the other smoking categories. The “Never Smoked” (“1”) and “Formerly Smoked” (“0”) groups also contain stroke cases, while the “Unknown” category (“3”) shows relatively fewer observations. These results suggest that smoking behavior may be related to stroke occurrence within the dataset.

Correlation Heatmap

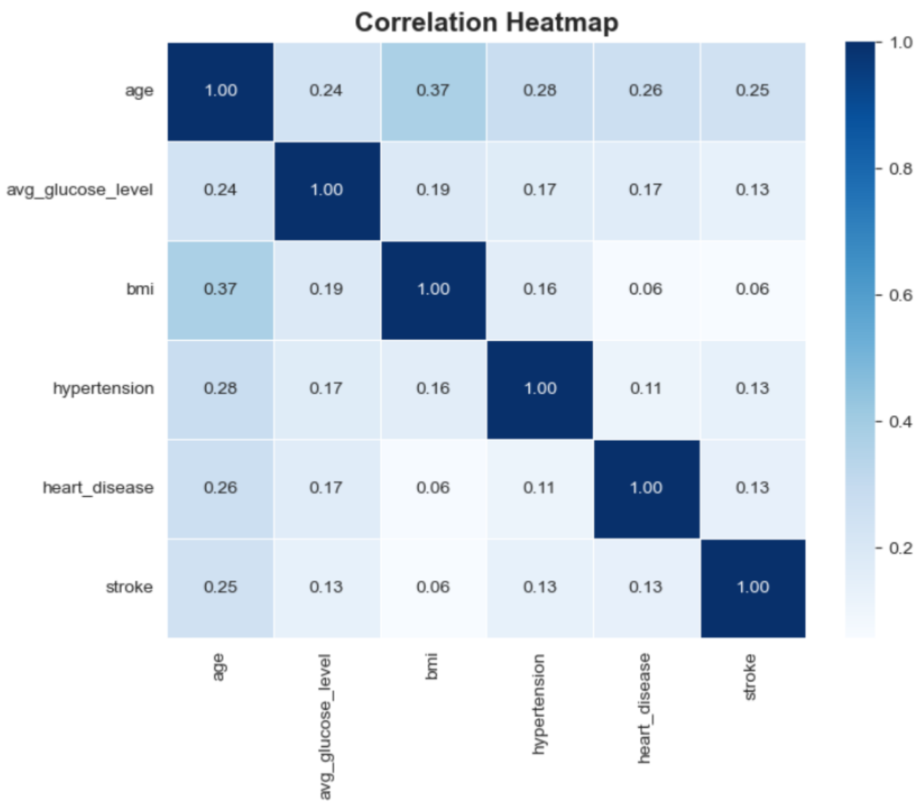


Figure 5.22: Correlation Heatmap of Numerical Variables

Figure 5.22 presents the correlation relationships among the numerical variables in the dataset. Age shows the strongest positive correlation with BMI (($r = 0.37$)) and moderate positive relationships with hypertension, heart disease, and stroke occurrence. The stroke variable demonstrates relatively weak positive correlations with age, hypertension, heart disease, and average glucose level, suggesting that these factors may contribute to stroke risk. Overall, no strong multicollinearity is observed among the predictor variables.

06. Advanced Analysis

This section presents the advanced analysis conducted to develop predictive models for identifying individuals at risk of stroke occurrence. Machine learning techniques were used because they can effectively identify patterns and relationships within healthcare data. In this study, Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) classification models were implemented and compared to determine the most suitable model for stroke prediction.

Before model development, the dataset was preprocessed using encoding, feature scaling, and class balancing with the Synthetic Minority Oversampling Technique (SMOTE). The dataset was then divided into training and testing datasets to evaluate model performance.

Logistic Regression Model

Logistic Regression was applied to predict stroke occurrence using demographic, medical, and lifestyle-related variables. This model was selected because it is simple, interpretable, and suitable for binary classification problems where the outcome variable has two categories, namely stroke occurrence and non-stroke occurrence. In healthcare research, Logistic Regression is widely used because it estimates the probability of disease occurrence and helps identify the influence of each predictor variable on the target variable.

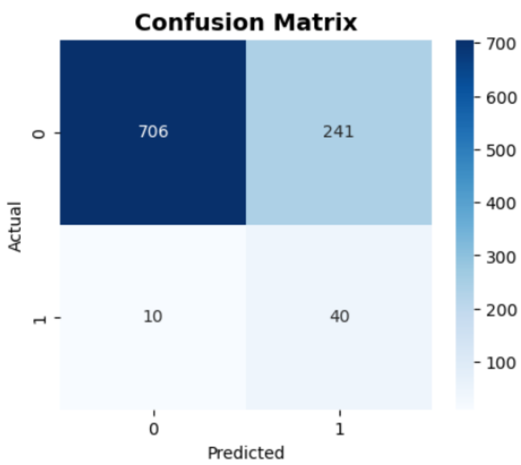
The fitted Logistic Regression equation is:

$$\log\left(\frac{P(Stroke)}{1-P(Stroke)}\right) = -1.246+1.987(age)+0.186(avg_{glucoselevel})+0.06(bmi)-0.059(gender_{male}) +0.17(hypertension_{Yes})+0.017(heartdisease_{Yes})+0.344(worktype_{Private}) +0.092(worktype_{selfEmployed})+0.77(worktype_{Children})+0.005(Residencetype_{Urban})- 0.157(Smokingstatus_{Never\ smoked})+0.094(Smokingstatus_{smokes})+0.131(\text{Ever\_married}_{\text{Yes}})+ 0.023(Smokingstatus_{\text{formerly smoked}})$$

Interpretation of Model Coefficients

- Intercept (-1.246):
  Represents the baseline log-odds of stroke occurrence when all predictor variables are zero.
- Age (1.987):
  Age has the largest positive coefficient in the model, indicating that increasing age substantially increases the probability of stroke occurrence. This suggests that age is the strongest predictor of stroke risk.

- Hypertension (0.170):
  Individuals with hypertension have a higher likelihood of experiencing stroke compared to individuals without hypertension.
- Heart Disease (0.017):
  Heart disease shows a small positive effect on stroke occurrence, indicating a slight increase in stroke risk.
- Average Glucose Level (0.186):
  Higher glucose levels are positively associated with stroke occurrence, suggesting that glucose level is an important medical risk factor.
- BMI (0.060):
  BMI shows a slight positive relationship with stroke risk, indicating that higher BMI may contribute to stroke occurrence.
- Gender (Male = 1) (-0.059):
  Male individuals show a slightly lower predicted stroke risk compared to females, although the effect is relatively small.
- Ever Married (Yes = 1) (0.131):
  Individuals who have been married show a slightly higher probability of stroke occurrence compared to unmarried individuals.
- Work Type Variables:
  The positive coefficients for private employment, self-employment, and children categories indicate varying levels of contribution to stroke prediction compared to the reference category.
- Residence Type (Urban = 1) (0.005):
  Residence type has a very small positive effect, indicating minimal influence on stroke occurrence.
- Smoking Status Variables:
  Current smokers show a slight positive relationship with stroke occurrence, while individuals who never smoked show a small negative relationship compared to the reference category.



Model Performance Interpretation

```
Accuracy: 0.7482447342026078
Precision: 0.1423487544483986
Recall: 0.8
F1 Score: 0.24169184290030213
ROC-AUC: 0.8389862724392819

Classification Report
              precision    recall  f1-score   support

           0       0.99      0.75      0.85       947
           1       0.14      0.80      0.24        50

    accuracy                           0.75       997
   macro avg       0.56      0.77      0.55       997
weighted avg       0.94      0.75      0.82       997
```

Figure 6.1: Confusion Matrix for the Logistic Regression Model

Figure 6.2 - Classification Report of the Logistic Regression model

The Logistic Regression model achieved an accuracy of 74.82%, indicating that approximately 75% of observations were correctly classified. The recall value of 0.80 shows that the model successfully identified 80% of actual stroke cases, which is important in healthcare applications where correctly detecting stroke patients is a priority.

The precision value of 0.14 indicates that some non-stroke cases were incorrectly classified as stroke cases. The F1-score of 0.24 reflects the imbalance between precision and recall. However, the ROC-AUC value of 0.84 demonstrates that the model has good overall ability to distinguish between stroke and non-stroke cases.

The confusion matrix further explains the classification performance of the model. The model correctly classified 706 non-stroke cases and 40 stroke cases. However, 241 non-stroke cases were incorrectly predicted as stroke cases, while 10 actual stroke cases were incorrectly classified as non-stroke cases. These findings indicate that the model prioritizes identifying stroke cases, even though it generates a relatively high number of false positive predictions.

Overall, the Logistic Regression model demonstrated moderate predictive performance and strong sensitivity in identifying stroke cases, making it useful as an initial screening model for stroke risk prediction.

Random Forest Model

The Random Forest model was applied to predict stroke occurrence using demographic, medical, and lifestyle-related variables. Random Forest was selected because it can effectively handle nonlinear relationships, reduce overfitting, and improve predictive accuracy by combining multiple decision trees. In healthcare datasets with complex patterns and imbalanced classes, Random Forest is often useful for identifying important predictors and improving classification performance.

Unlike Logistic Regression, Random Forest does not produce a single mathematical equation because predictions are generated using an ensemble of multiple decision trees. The final prediction is determined through majority voting among the trees in the forest.

Model Performance Interpretation

```
Accuracy: 0.8966900702106319
Precision: 0.09230769230769231
Recall: 0.12
F1 Score: 0.10434782608695652
ROC-AUC: 0.8096409714889123

Classification Report
              precision    recall  f1-score   support

           0       0.95      0.94      0.95       947
           1       0.09      0.12      0.10        50

    accuracy                           0.90       997
   macro avg       0.52      0.53      0.52       997
weighted avg       0.91      0.90      0.90       997
```

Figure 6.3: Classification report of the Random Forest Model

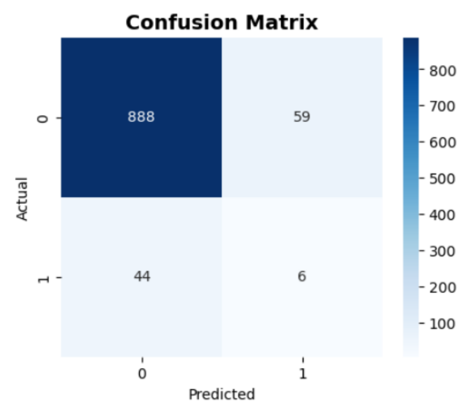


Figure 6.4: Confusion Matrix for the Random Forest Model

The Random Forest model achieved an accuracy of 89.67%, indicating strong overall classification performance. However, the recall value of 0.12 shows that the model identified only a small proportion of actual stroke cases. The precision (0.09) and F1-score (0.10) also indicate limited performance for the minority stroke class, while the ROC-AUC value of 0.81 demonstrates acceptable classification ability.

The confusion matrix shows that the model correctly classified 888 non-stroke cases and 6 stroke cases. However, 59 non-stroke cases were incorrectly predicted as stroke cases, while 44 actual stroke cases were missed. These results suggest that the model performed well in identifying non-stroke cases but struggled to accurately detect stroke cases due to class imbalance in the dataset.

Feature Importance Interpretation

Random Forest Feature Importance

	Feature	Importance
0	age	0.345988
3	avg_glucose_level	0.144361
4	bmi	0.140843
6	ever_married_Yes	0.081338
10	Residence_type_Urban	0.052328
5	gender_Male	0.050007
1	hypertension	0.033802
7	work_type_Private	0.033738
12	smoking_status_never smoked	0.031310
11	smoking_status_formerly smoked	0.019128
8	work_type_Self-employed	0.018899
13	smoking_status_smokes	0.017494
2	heart_disease	0.015467
9	work_type_children	0.015297

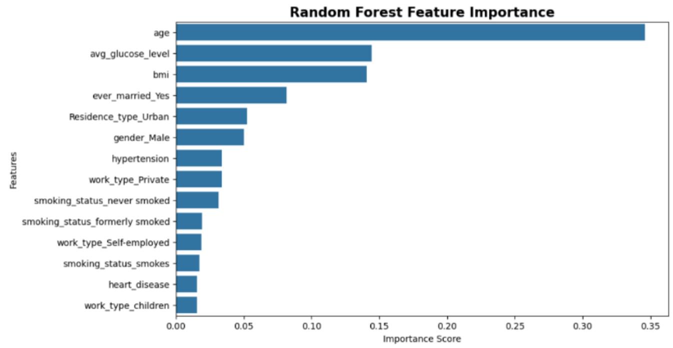


Table 6.1:Feature Importance Scores of the Random Forest Model

Figure 6.5:Feature Importance Plot of the Random Forest Model

The Random Forest feature importance results indicate that age was the most influential predictor of stroke occurrence. Average glucose level and BMI also showed strong contributions to the prediction process. Variables such as hypertension, smoking status, and work type had moderate influence, while heart disease showed relatively lower importance in the model.

K-Nearest Neighbors (KNN) Model

The K-Nearest Neighbors (KNN) model was applied to predict stroke occurrence using demographic, medical, and lifestyle-related variables. KNN was selected because it is a simple and effective non-parametric classification technique that classifies observations based on the similarity of neighboring data points. The model is useful for identifying hidden patterns in medical datasets without making assumptions about data distribution.

Model Performance Interpretation

```
Accuracy: 0.8144433299899699
Precision: 0.10526315789473684
Recall: 0.36
F1 Score: 0.16289592760180996
ROC-AUC: 0.6429355860612461

Classification Report
              precision    recall  f1-score   support

           0       0.96      0.84      0.90       947
           1       0.11      0.36      0.16        50

    accuracy                           0.81       997
   macro avg       0.53      0.60      0.53       997
weighted avg       0.92      0.81      0.86       997
```

Figure 6.6:Classification Performance Metrics of the KNN Model

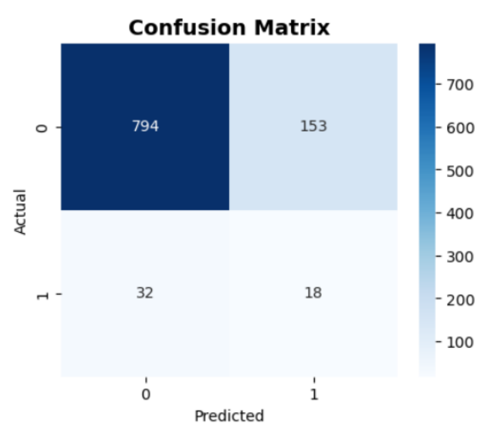


Figure 6.7:Confusion matrix of the KNN Model

The KNN model achieved an accuracy of 81.44%, indicating good overall classification performance. The recall value of 0.36 shows that the model identified 36% of actual stroke cases, while the precision value of 0.11 indicates that several non-stroke cases were incorrectly predicted as stroke cases. The ROC-AUC value of 0.64 suggests relatively weaker classification ability compared to the other models.

The confusion matrix shows that the model correctly classified 794 non-stroke cases and 18 stroke cases, while 153 non-stroke cases were incorrectly predicted as stroke cases and 32 stroke cases were missed. Overall, the KNN model demonstrated moderate predictive performance and provided a balance between identifying stroke and non-stroke cases. However, its performance was lower than Logistic Regression in detecting stroke cases and lower than Random Forest in overall accuracy.

ROC Curve Comparison

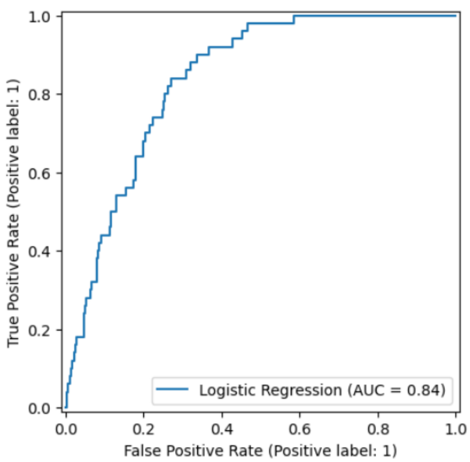


Figure 6.8:ROC Curve for the Logistic Regression Model

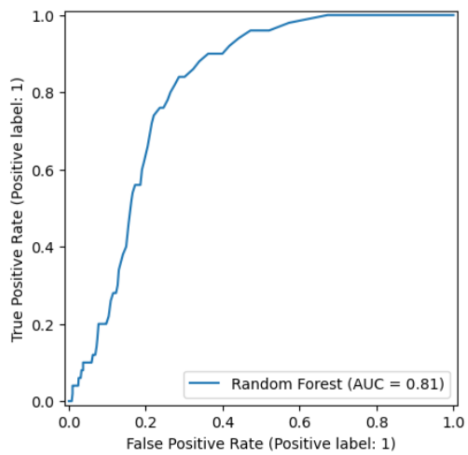


Figure 6.9:ROC Curve for the Random Forest Model

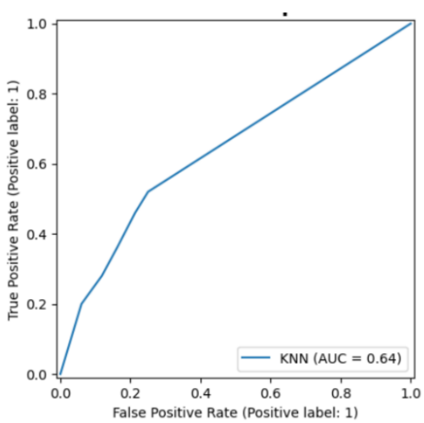


Figure 6.10:ROC Curve for the KNN Model

The ROC curve comparison indicates that the Logistic Regression model achieved the highest classification performance with an AUC value of 0.84, showing better ability to distinguish between stroke and non-stroke cases. The Random Forest model also demonstrated good classification performance with an AUC value of 0.81. In contrast, the K-Nearest Neighbors (KNN) model showed comparatively weaker performance with an AUC value of 0.64. Overall, Logistic Regression performed best among the three models based on ROC-AUC results.

Model Comparison

	Model	Accuracy
0	Logistic Regression	0.748245
1	Random Forest	0.896690
2	KNN	0.814443

Tabel 6.2: Comparison of Classification Accuracy Among Machine Learning Models

The model comparison results indicate that the Random Forest model achieved the highest accuracy (89.67%), demonstrating the strongest overall classification performance among the three models. The K-Nearest Neighbors (KNN) model achieved moderate accuracy (81.44%), while Logistic Regression showed the lowest accuracy (74.82%). Overall, Random Forest outperformed the other models in terms of overall predictive accuracy for stroke prediction.

07. General Discussion and Conclusion

General Discussion

- This study aimed to predict the likelihood of brain stroke occurrence using demographic, medical, and lifestyle-related variables obtained from a Kaggle dataset. Three machine learning classification models Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest were implemented and compared based on their predictive performance.
- Exploratory Data Analysis (EDA) revealed that age, hypertension, heart disease, average glucose level, and BMI were important factors associated with stroke occurrence. The dataset also showed a severe class imbalance, with substantially fewer stroke cases compared to non-stroke cases.
- Logistic Regression demonstrated moderate predictive performance with an accuracy of 74.82% and the highest ROC-AUC value of 0.84 among the three models. The model successfully identified most stroke cases, achieving a recall value of 0.80. However, the model produced a relatively high number of false positive predictions, resulting in low precision.
- The K-Nearest Neighbors (KNN) model achieved an accuracy of 81.44%, showing better overall accuracy than Logistic Regression. However, its recall value of 0.36 indicated limited effectiveness in detecting actual stroke cases. The ROC-AUC value of 0.64 also suggested weaker classification performance compared to the other models.
- The Random Forest model achieved the highest overall accuracy of 89.67%, indicating strong performance in classifying non-stroke cases. Feature importance analysis further identified age, average glucose level, and BMI as the most influential predictors of stroke occurrence. However, despite its high accuracy, the model achieved a low recall value of 0.12, showing difficulty in correctly identifying stroke-positive patients.
- The results indicate that high model accuracy alone is insufficient in medical prediction problems involving imbalanced datasets. Models with high accuracy may still fail to identify minority stroke cases effectively.
- The findings also demonstrate that age consistently emerged as the strongest predictor across all models, followed by average glucose level, hypertension, and BMI. These results align with previous studies highlighting the importance of these variables in stroke prediction.
- Although SMOTE balancing techniques were applied to reduce class imbalance, all models still showed challenges in accurately detecting stroke-positive cases. This

suggests that stroke prediction remains a complex classification problem requiring more advanced balancing and optimization techniques.

Conclusion

- This study successfully applied Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN) models to predict stroke occurrence using demographic and clinical variables.
- Among the three models, Random Forest achieved the highest overall classification accuracy, while Logistic Regression demonstrated the best ability to distinguish between stroke and non-stroke cases based on ROC-AUC performance.
- The study identified age, average glucose level, hypertension, and BMI as the most important factors influencing stroke prediction.
- Despite the application of SMOTE for class balancing, the models still experienced difficulty in accurately identifying stroke-positive cases due to the imbalanced nature of the dataset.
- Overall, the findings suggest that machine learning techniques can support early stroke risk prediction and assist healthcare professionals in identifying high-risk individuals. However, further improvements using advanced resampling techniques, hyperparameter tuning, or deep learning approaches may enhance predictive performance in future studies.

08. References

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